# Development and validation of a mold making process using clay for reverse engineering of Pelton runners

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**Abstract.** A Pelton runner during its operational period incurs various forms of wear, resulting in a distortion of the geometry. During the repair and maintenance of these runner buckets, it is desirable to obtain the original geometry to retain the efficiency of the turbine. In many powerplants, the drawings of the buckets are usually not provided, which may cause discrepancies between actual and repaired geometry. This paper aims to investigate the prospect of using molding and casting techniques to develop a 3D shape of the available runner. A bucket of a 26 kW Pelton runner is studied, and a mesh model is created for reference. For the molding process, clay molding technique is used. Buckets molded from this process is casted in fiberglass and gelcoat. The buckets are scanned and thus obtained mesh models are compared with the mesh model of the reference bucket. It was observed that the bucket molded from the clay molding process and cast in mixture of fiberglass and gelcoat has the results with a mean of -0.03 mm and a standard deviation of 0.43 mm, which depicts the cluttering of data around the targeted value. Thus, clay molding process can be used for creating a mold, while mixture of fiberglass and gelcoat can be used as the casting material to replicate its geometry. This approximation can be further analysed by incorporating the data obtained from the results. Further detailing can be done to increase the efficiency of the process used. Moreover, if the process developed in this paper is commercialized this technique would significantly reduce the cost and time for redesign, research, and even maintenance of Pelton runners.

***Keywords:*** *cloud compare, 3D scanning, 3D modelling, fiberglass molding, mesh editing*

## 1. Introduction

Despite having great potential in hydropower production, Nepal suffers a major drawback in sediment management[1–3]. At the intakes, coarse sediments like gravel and sand can be kept out of the river flows, but it is not economically possible to keep silt and clay particles out of the water going into hydraulic turbines. Even such tiny particles in Pelton turbines, especially those installed in high-head HPPs, inflict significant erosion damage to the runners' buckets. Critical parts like the impeller, turbine blade, and casing were found to have cavitation penetration rates as high as 10 mm/year [4]. According to the reports on hydro abrasive erosion in turbine components[5], the turbine runner often loses 200 kg worth of metal over the course of a few years of operation. A significant amount of metal loss identifies cavitation erosion as a major issue.

Hydraulic turbines that experience hydro-abrasive erosion have lower turbine efficiency and experience cavitation. Particularly important for turbine effectiveness is the splitter[6], which is placed in the centre plane of each bucket and is in charge of ensuring smooth jet bifurcation. Reduced HPP efficiency and eventual HPPs failures result in lower electricity production and revenue. The cost effectiveness of HPPs is harmed along with increased expenditures for maintenance and replacements[7–9]. Cavitation-led erosion or pitting may reach a depth of 40 mm[10] or more, at which a hydro turbine runner is deemed unsafe for use and is often removed from service for maintenance.

Complete damages cannot be fully prevented even using special hard-coatings[11] and enhancing the turbine efficiency largely depends upon Pelton turbine's primary geometric bucket parameters-such as length, width, and depth[12] which will also be affected by the erosion during its use and eventually needs periodic repair and welding making repair and welding a viable option which is to date, the most successful method of repairing cavitation damage on hydraulic turbines[13].

Lester A. Pelton, the Pelton turbine's inventor, patented the device in the United States in October 1880, but not all of the designs have been resolved [14]. There is still a dearth of information in the public domain on the turbine bucket design process and the ideal geometric form. The intense competition in the hydroelectricity industry, which keeps new information a secret, contributes to the situation as stated above[15]. Because the information is protected as a confidential corporate secret, there are relatively few details on the design and study of the turbine[16].

Before proceeding with repairs, it's important to weigh the available options for the task. There are two primary approaches to consider: one involves restoring the runner to its original profiles, while the other entails modifying the runner to eliminate or minimize the underlying cause(s) of the damage[13]. Restoring the runner to its original profile is often seen as the simpler and more direct approach. In many powerplants, the drawings of the buckets are usually not provided[17], which may cause discrepancies between actual and repaired geometry.

This paper aims to investigate the prospect of using molding and casting techniques to develop a 3D digital model of the runner at the beginning of its operation through Reverse Engineering (RE) techniques and comparing the result obtained from molding and casting techniques with reference model. This paper is intended to have impact in the repair and maintenance of the Pelton bucket having a reference digital copy of the original profile which will aid in retaining the original efficiency of the runner.

## 2. Material and Methods

### *2.1 Model*

Before proceeding validation to assure that mold do accurately provide the original shapes and retain them. To do so a smaller version of a runner to compare the original scan of its dismantled bucket and a mold scan of the bucket making it easier to validate that mold indeed does give similar results to its original profile. To accomplish this, Pelton turbine with 18 buckets which has a capacity of 26 KW with design flow rate of 50 litres per second and design head 100 meters was used.

Figure 1. Reference model for scanning

*2.2 Reference Model Generation*

The Pelton bucket was firmly affixed to the model, prompting the removal of the runner for easier access. To facilitate effective 3D scanning, the bucket's red paint was carefully stripped using Dimethylformamide (DMF). This process ensured optimal conditions for scanning. Markers were strategically positioned on the model to ensure seamless transitions from one point to another during the scanning process. The 3D scanning was conducted utilizing the Shining 3D-Ein Scan HX[18], which was accessible at the Design Lab of Kathmandu University. The scanning specifications are outlined in Table 1. The scanning procedure encompassed four calibration types: standard calibration, white balance calibration, laser calibration, and quick calibration.

**Table 1.** Shining 3D-Ein Scan HX Specifications[18]

| | **Laser Scan** | **Rapid Scan** |
|---|---|---|
| **Scan** | 300mm-4m objects | |
| **Light source** | Laser | LED |
| **Accuracy** | 0.04mm | 0.05mm |
| **Resolution** | 0.05mm-3mm | 0.25-3mm |
| **Alignment** | Markers | Markers/Features/Hybrid/Texture |
| **Scanning Speed** | 55FPS | 15FPS (non-texture);10FPS (texture) |
| **Texture Scan** | No | Yes |

Laser scanning was performed on the model using medium mesh settings and low-resolution settings, set at 1.0 mm to capture the point clouds. The captured point clouds underwent an editing process, eliminating unnecessary data. This procedure was repeated twice from different perspectives, recognizing that capturing all model data from a single viewpoint was not possible. The captured point clouds were then aligned, using one view as a fixed reference and the other two as floating views. This alignment process facilitated the creation of a comprehensive mesh for the model. The resulting mesh underwent further editing to remove the markers and automatically fill the voids generated from marker removal. The mesh was then refined through smoothing and sharpening techniques, incorporating simplification to optimize polygon count, file size, and data complexity. The refined mesh was exported in STL format. The final mesh model, obtained after scanning, is shown in Figure 2.

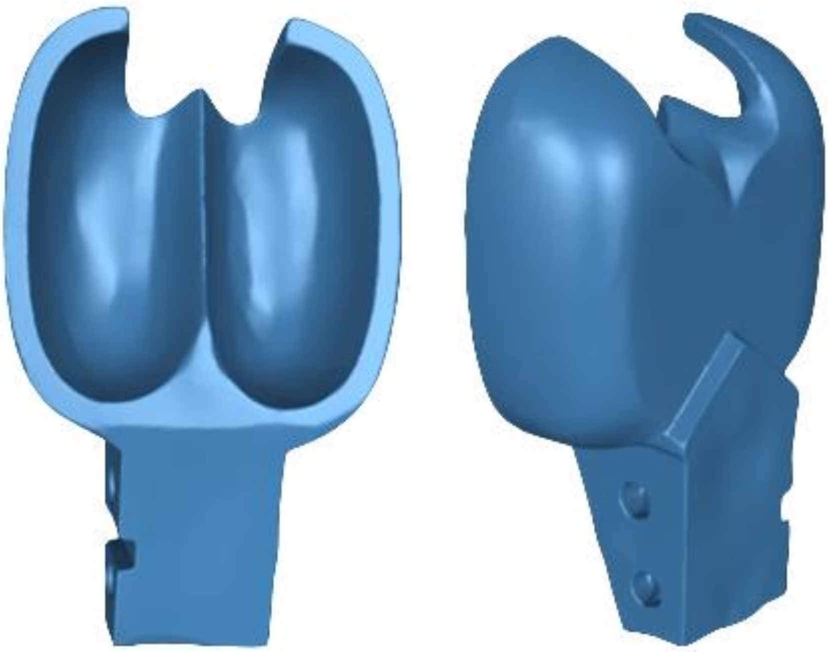

Figure 2. Mesh obtained from 3D scanning

]*2.3 Geometry Extraction*

*2.3.1 Molding*

The clay molding technique was used to replicate the geometry of a reference model, specifically for the fabrication of a Pelton bucket. The process began with cleaning the bucket to ensure that all surfaces maintained their original imprint and were free of impurities or residues. A thin layer of oil and Petroleum Jelly was applied to the surface of the bucket to aid in the formation of the mold. This layer functioned as a separator, facilitating the separation of the mold material from the model upon completion of the mold.

Pottery clay, as specified by properties listed in Table 2, was pressed against the bucket model to capture its shape and details. The clay was manipulated to form a mold space that mirrored the geometry of the bucket. The resulting mold would influence the precision and quality of the final casting. A two-piece mold was fabricated, with one mold section for the front face and another for the second face. This design facilitated the extraction of the molded object.

**Table 2.** Physical Property of Pottery Clay

| Property | Numeric Value |
|---|---|
| **Moisture Content** | 20-25% |
| **Shrinkage** | 2-4% |
| **Firing Temperature** | 980-1350$^{o}$ C |
| **Porosity** | 5-25% |
| **Density** | 1.5-2.5 $\frac{gm}{cm^3}$ |
| **Thermal Expansion** | $5E-06/^{0}$C |

Subsequent to the formation of the mold, a curing process is initiated to ensure the mould's structural integrity and stability. During this period, the clay is allowed to set and solidify, resulting in a mold that represents the original bucket’s shape. Upon completion of the curing process, the clay is removed from the mould with emphasis given to prevent any residue or remnants of the clay from adhering to the model. This extraction process is essential to maintaining the accuracy of the molded shape and facilitates a smooth demoulding process.

**Figure 3.** Mold obtained from Clay Molding Process

*2.3.2 Casting*

Two moulds were obtained through the clay molding process, after which the casting phase began. A layer of Vaseline and oil was applied to act as a separator, ensuring that the casting material could be separated from the moulds.

In the casting procedure, a gelcoat was formulated using a composition of unsaturated polyester resin, a white pigment, and cobalt. The cobalt served as an accelerator for the resin's settling. The gelcoat was applied to the surface of the moulds, providing coverage and eliminating any bubbles that formed during this application. After the application, the gelcoat was allowed to dry. It was considered ready for the next step when it no longer adhered to the hand upon touch.

After the initial gelcoat layer dried, a second layer of gelcoat was applied. Fiberglass was then added to the moist surface of the mold. To encapsulate the fiberglass, a layer of gelcoat was applied, covering and submerging the fiberglass material. The application ensured that the fiberglass was integrated into the mold and suppressed within the gelcoat. The casts, embedded with the gelcoat and fiberglass, were left to dry. This drying period allowed the gelcoat and fiberglass to solidify, contributing to the final product's structural stability and integrity. Once dried, two casts were extracted from their respective moulds, preserving their shapes and details.

To complete the Pelton bucket, these two individual casts were joined together using a putty made of polyester resin. Polyester resin was chosen for its drying properties, facilitating a quick and efficient joining process. The putty was applied, ensuring a bond between the two casts, resulting in a Pelton bucket that was ready for use.

A final cast was obtained, as shown in figure 4. A smooth surface can be observed in the figure, which was a result of grinding. A misalignment of the lip can also be observed in the figure, which was also a result of grinding.

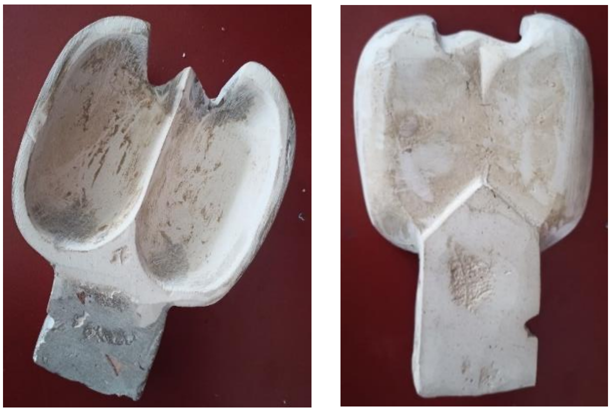

Figure 4. Final Cast of the Pelton Bucket

*2.4 Comparison of Casted model with Reference model*

The same process of scanning was used to obtain the mesh profile of the cast thus obtained. Then the two mesh profiles were compared as their mesh-to-mesh distances after their fine alignment, were measured by assigning the mesh profile of the original Pelton Runners bucket as a reference geometry. In CloudCompare the two profiles that are to be compared are aligned roughly by using the interactive transformation tool, then they are finely registered using the fine registration tool (ICP), this tool uses the iterative closest point algorithm for fine registration of the point cloud. It is used to refine the coarse alignment for obtaining the best fit for aligning the meshes. The CC works with 32 bits floating values, a $e^{-8}$ threshold is already the computation accuracy limit[19]. Thus, the RMS difference between the two iterations didn't decrease in registration of the two meshes in this project.

The results obtained in CC are based on two parameters, one is the octree creation and the other is the C2M distances. Octrees are used to partition a three-dimensional space by recursiely subdividing it into eight octants. In a point region(octree), the nodes store an explicit three-dimensional point, which is the center of the subdivision for that node[19].

For mesh-to-mesh comparison the mesh-mesh distance tool is used, which provides the deviation of the point in the mesh to be compared with the reference mesh. The C2M signed distances are the Cloud to mesh distances, the distance of the mesh to be compared relative to the reference mesh.

## 3. Results

The results obtained from CloudCompare showed evident deviations in the red region of figure 5 and 6 at the lip and towards the superficial surface of the bucket when compared with the reference mesh. These results were expected due to the grinding used after the cast was prepared.

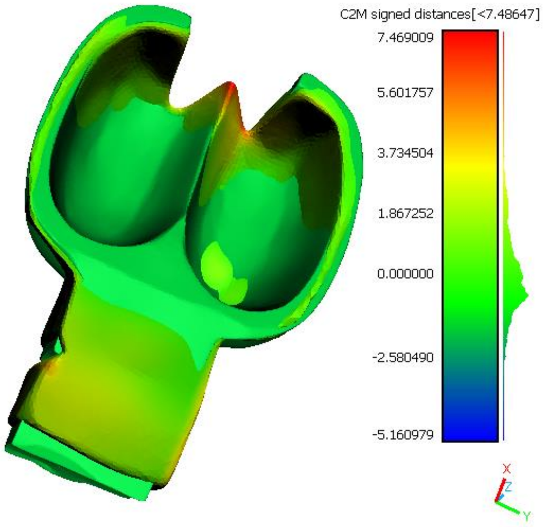


**Figure 5.** Front view of the two aligned meshes

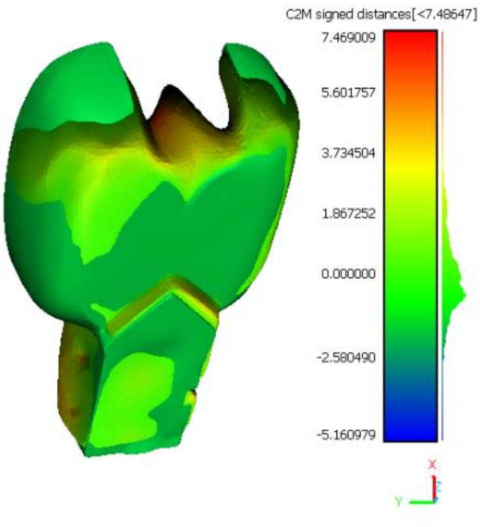


**Figure 6.** Rear view of the two aligned meshes

The heat map located to the right of each of the figures 5 and 6 shows the C2M distances between the octrees of the reference mesh and the mesh of the cast after fine alignment. Furthermore, this result can be displayed in a histogram to further understand the implications of the data.

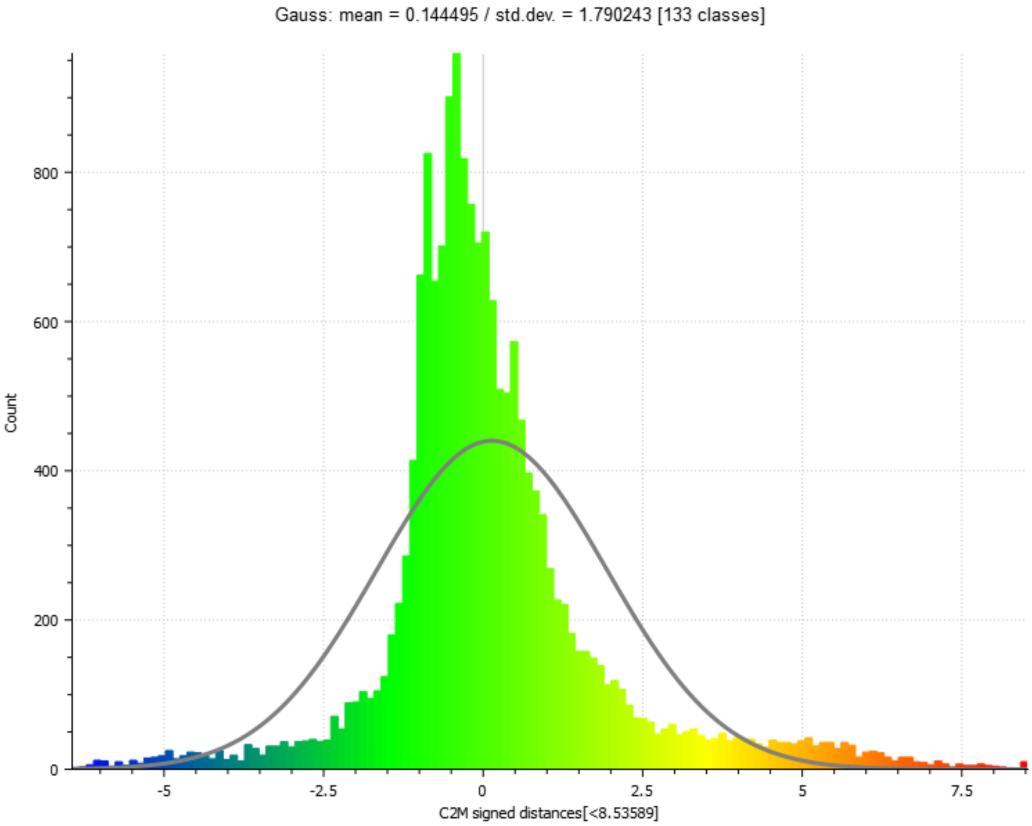


**Figure 7.** Histogram of the C2M signed distances obtained from CC

Figure 7, which was also extracted from the comparison results from CC, shows the number of deviations in the y-axis. The negative sign indicates shrinkage in the geometry, while the positive sign indicates an increase in length of the octree of mesh model of the cast as compared to the reference mesh, while in the x-axis are the C2M signed distances from the octrees of reference mesh to the mesh of the cast within a range of -6.440 to 8.5 mm.

## 4. Conclusion

In the Gaussian distribution of the deviations, the calculated mean value was found to be 0.14495 mm. This central tendency suggests that the majority of the data points are clustered around this value. Additionally, the standard deviation was found to be approximately 1.790243 mm, indicating the spread of the data around the mean. This shows the overall variation in the geometry of the cast with reference to the original bucket of the Pelton runner. Furthermore, the heat map obtained from Cloud Compare enables the calculation of deviations in desired points of the geometry. The results shows that the process can be replicable to other areas of research where this process and technique can be used for reverse engineering. This technique can help analyse and optimize various aspects of the bucket, such as its shape and size, with the aim of improving the overall efficiency of the turbine. This optimization can lead to higher energy conversion rates and more efficient power generation.

REFERENCES


[1] Thapa B, Shrestha R, Dhakal P and Thapa B S 2005 Problems of Nepalese hydropower projects due to suspended sediments *Aquat Ecosyst Health Manag* **8** 251–7

[2] Ge X, Sun J, Chu D, Liu J, Zhou Y, Zhang H, Zhang L, Chen H, Kan K, Binama M and Zheng Y 2023 Sediment Erosion on Pelton Turbines: A Review *Chinese Journal of Mechanical Engineering (English Edition)* **36**

[3] Andermann C, Bonnet S, Crave A, Davy P, Longuevergne L and Gloaguen R 2012 Sediment transfer and the hydrological cycle of Himalayan rivers in Nepal *Comptes Rendus - Geoscience* **344** 627–35

[4] Rai A K, Kumar A, Staubli T and Yexiang X 2013 *Interpretation and application of the hydro-abrasive erosion model from IEC 62364 (2013) for Pelton turbines 2*

[5] Singh R, Tiwari S K and Mishra S K 2012 Cavitation erosion in hydraulic turbine components and mitigation by coatings: Current status and future needs *J Mater Eng Perform* **21** 1539–51

[6] Duan C G and Karelin V Y 2003 Abrasive Erosion and Corrosion of Hydraulic Machinery **2**

[7] Thapa B S, Thapa B and Dahlhaug O G 2012 Empirical modelling of sediment erosion in Francis turbines *Energy* **41** 386–91

[8] Thesis *HYDRO-ABRASIVE EROSION OF PELTON TURBINES*

[9] Felix D 2017 Experimental Investigation On Suspended Sediment, Hydro-Abrasive Erosion And Efficiency Reductions Of Coated Pelton Turbines

[10] Nikitenko G I 1998 Studies to reduce cavitation erosion of the impellers in the hydro turbines at the Sayano-Shushenskoe Hydroelectric Plant *Power Technology and Engineering* **32** 570–3

[11] Zou H, Yan S, Shen T, Wang H, Li Y, Chen J, Meng Y, Men S, Zhang Z, Sui T and Lin B 2021 Efficiency of surface texturing in the reducing of wear for tests starting with initial point contact *Wear* **482–483** 203957

[12] Židonis A, Panagiotopoulos A, Aggidis G A, Anagnostopoulos J S and Papantonis D E 2015 Parametric optimisation of two Pelton turbine runner designs using CFD *Journal of Hydrodynamics* **27** 403–12

[13] Anon TURBINE REPAIR BUREAU OF RECLAMATION April 1989 By William Duncan Jr : Free Download, Borrow, and Streaming : Internet Archive

[14] Quaranta E and Trivedi C 2021 The state-of-art of design and research for Pelton turbine casing, weight estimation, counterpressure operation and scientific challenges *Heliyon* **7**

[15] Solemslie B W and Dahlhaug O G 2012 A reference Pelton turbine design *IOP Conf Ser Earth Environ Sci* **15** 032005

[16] Sabu S, Antony T A, George N J, Chandy A, Asst A and Jyothi A *Design and Modelling of a Pelton Wheel Bucket Theoretical Validation And Software Comparison*

[17] Zhang Z 2016 Pelton turbines *Pelton Turbines* 1–311

[18] Anon 2020 *SHINING 3D-EinScan HX 1.0.2.1-User Manual-September 2020 EinScan HX User Manual*

[19] Girardeau Daniel ICP - CloudCompareWiki